

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.5: AREA OF POLYGONS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we calculate the area of any polygon, no matter how many sides or how irregular, so that we can fairly divide the Kericho community land plot and solve other real Kenyan land and design problems? Answer each section in full sentences. Show all working for calculations, with units. Draw labelled diagrams where they help. Use what you observed and measured in Lessons 1 to 8: the Kericho plot, the Kisumu solar panel, the Mombasa courtyard and the Nakuru school shamba. Give a reason for every claim.

1. The Kericho plot: why one formula is not enough

A village near Kericho has a seven-sided plot to share among three farming families. Nobody at the meeting could find its area with one formula they remembered. (a) Explain why no single standard formula such as the one for a rectangle works for this plot. (b) Describe the method you used to turn the problem into one you can solve. Say how many shapes the plot is split into and how you draw the lines. (c) Explain why the height used in $A = \frac{1}{2}bh$ must be the perpendicular height and not a slanted side.

Write your answer here:

2. Tools for pieces of the plot: quadrilaterals and the hidden height

Some pieces of the plot are quadrilaterals, and some triangles have a height that cannot be measured. (a) Write the area formulae you derived for the parallelogram, rhombus, trapezium and kite. For the parallelogram, explain how cutting and rearranging it shows its formula. (b) In Kisumu, a triangular solar panel has two sides of 5 m and 7 m with an included angle of 58° . Explain why $A = \frac{1}{2}bh$ is hard to use here. (c) Derive the formula $A = \frac{1}{2}ab \sin C$ from $A = \frac{1}{2}bh$, then calculate the area of the panel. Give your answer to 2 decimal places.

Write your answer here:

3. Checking with a regular polygon and a scale diagram

(a) The Mombasa Old Town courtyard is a regular octagon with side 3 m. Explain how drawing lines from the centre O to each vertex lets you find its area, and state the central angle of each triangle. (b) Calculate the area using the direct formula $A = 2(1 + \sqrt{2})s^2$, to 2 decimal places. (c) Explain why the formula and the decomposition method must agree. (d) On the Nakuru school shamba diagram, 1 cm² represents 16 m². One triangle on the diagram has an area of 3 cm². Find its real area and explain why the conversion for area uses 16 rather than the conversion factor for lengths.

Write your answer here:

4. Dividing the Kericho plot fairly

The Chepkemoi, Mwangi and Otieno families each need an equal area of the Kericho plot. (a) Explain how you would find the total area of the plot. Describe how composite shapes can be handled with both adding and subtracting. (b) Explain how the three families' shares are worked out from the total. (c) Two groups split the plot in different ways and get totals that differ slightly. Give two possible reasons for the difference. State what it would mean if their methods were both correct and carried out accurately.

Write your answer here:

5. Answering the driving question

Write a short final answer to the driving question. (a) Explain how you can calculate the area of any polygon, no matter how many sides or how irregular. Include how many triangles an n-sided polygon is split into from one vertex. (b) Give one real Kenyan situation from our lessons where polygon area is used and explain why an accurate area matters in it. (c) Compare your first estimate of the Kericho plot in Lesson 1 with how you now solve the problem. Describe what you now understand better.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Decomposition method and the (n – 2) triangle idea	Clearly explains drawing diagonals from one vertex, states (n – 2) triangles correctly for the heptagon and other polygons, and explains that the total is the sum of the triangle areas.	Describes splitting into triangles and summing areas, with the number of triangles mostly correct or the reasoning partly explained.	Mentions shapes but does not explain how the polygon is split or how the parts combine into the total.
Use of formulae and calculation accuracy	Recalls all formulae correctly, derives $A = \frac{1}{2}ab \sin C$, and calculates 14.84 m ² , 43.46 m ² and 48 m ² with correct working and units.	Uses most formulae correctly and gets most calculations right, with minor slips in rounding or units.	Formulae are missing or confused (for example slant height instead of perpendicular height) and calculations are incomplete or incorrect.
Reasoning and links between methods	Explains why the perpendicular height is needed, why the octagon formula and decomposition agree, why area scale factors are squared, and why different valid decompositions give the same area.	Gives reasons for most of these ideas, though some explanations are brief or incomplete.	States answers without reasons or does not connect methods to each other.
Application to the Kericho land-division problem and Kenyan contexts	Clearly explains finding the total area, dividing it into three equal shares for the families, handling composite shapes by adding and subtracting, and the real-life importance of accurate area.	Explains how the plot is divided and gives a relevant Kenyan example, with some gaps in detail.	Gives a vague or incomplete explanation of the land division and little or no link to real contexts.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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